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Dear Dr. Öpik:

Please consider our paper, entitled ‘The problem and promise of modeling plant leafout under warmer winters,’ as a viewpoint for *New Phytologist*.

What hypotheses or questions does this work address?

Why do current models of leafout timing with climate change make wildly different predictions for the same system and temperature regime? How can simple models from the 1970s still be equally predictive as newer models, despite apparent progress over the last decades? We argue progress has stalled because we do not understand how winter temperatures affect leafout. Models today estimate the effect of winter temperatures via an unobserved process called chilling. This has produced fundamental problems that the field has struggled to admit or address.

How does this work advance our current understanding of plant science?

Chilling is a fundamental concept in plant biology with implications for many fruit crops and how well forests can continue to sequester carbon with climate change (1; 2). Here we highlight major problems with the concept that result in hidden assumptions and obscured work-arounds in models and experiments, leading to highly uncertain and variable forecasts. We show how new molecular advances (e.g., 3; 4; 5; 6) may rule out certain common models, and suggest a path towards more mechanistic models and better experiments.

Why is this work important and timely?

Interest in chilling for forecasting the effects of climate change has led to myriad new models, but few insights. In this perspective we argue that major advances through molecular and other new approaches will be slow without a reckoning with the concept of chilling.

In this article we bring a cross-disciplinary and solutions-focused perspective on chilling, bridging across plant physiology, molecular biology, forest tree and crop modelers to highlight the problems that may limit the potential for rapid progress, and show how to fix them. We expect our paper will be controversial, especially from researchers who have focused strongly on modeling chilling (e.g., those working on the Dynamic Model) but believe our viewpoint will accelerate progress. We suggest reviewers on the following page.

Sincerely,

A handwritten signature in black ink, appearing to read 'Elizabeth Wolkovich'.

Elizabeth M Wolkovich
Professor of Forest & Conservation Sciences

References:

- [1] Keenan, T. F. *et al.* Net carbon uptake has increased through warming-induced changes in temperate forest phenology. *Nature Climate Change* **4**, 598–604 (2014).
- [2] Pörtner, H. O. *et al.* *Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* (Cambridge University Press, 2022).
- [3] Zhu, P., Lister, C. & Dean, C. Cold-induced arabidopsis frigida nuclear condensates for flc repression. *Nature* **599**, 657–661 (2021).
- [4] Azeez, A. *et al.* Early bud-break 1 and early bud-break 3 control resumption of poplar growth after winter dormancy. *Nature communications* **12**, 1123 (2021).
- [5] Pan, W. *et al.* Epigenetic silencing of callose synthase by VIL1 promotes bud-growth transition in lily bulbs. *Nature Plants* **9**, 1451–1467 (2023).
- [6] Cai, F. *et al.* Molecular regulation of bud dormancy in perennial plants. *Plant Growth Regulation* **102**, 1–11 (2024).

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